# Daniel Firoozabadi

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## Education/Academic Summary

**UC Berkeley - EECS - GPA: 3.7**

- **Relevant Coursework**: Data Structures & Algorithms; Machine Structures; Linear Algebra; Discrete Mathematics & Probability Theory; Signals & Systems; Structure & Interpretation of Computer Programs; Multivariable Calculus
- **Organizations:** AI Entrepreneurs at Berkeley - Builder Fellow, Adobe Campus Ambassador, Perplexity Strategist, Berkeley Entrepreneurship Hub, Skydeck
- **Expected Graduation:** Spring 2028

## Technical Skills

**Languages:** Python, Java, C/C++, JavaScript, TypeScript, SQL, MATLAB
**Frameworks/Tools:** TensorFlow, React, NumPy, Pandas, Node.js, AWS (Amplify, DynamoDB), Git, MySQL, PyVista, OpenMDAO

## Work/Research Experience

**Incoming TIP Software Engineering Intern (Summer 2026) - Capital One**

- Backend and infrastructure engineering for large-scale, production systems

**AI Researcher (Contract) - Capital One**
**August 2025 to December 2025**

- Automating internal AI workflows using Python-based APIs and the Gemini ADK to streamline financial operations and reduce repetitive manual review.
- Building multi-agent automation pipelines that leverage Gemini ADK context and intent models to improve decision reliability and workflow consistency.
- Collaborating with engineering and product teams to design, test, and deploy LLM-driven automation solutions that accelerate internal review cycles.

**Machine Learning Researcher - Intellikgraph Research & Consulting**
**August 2025 to December 2025**

- Developing confidence-aware explanation methods grounded in structured maritime knowledge graphs (AllegroGraph, SPARQL).
- Implementing Python-based simulation and uncertainty-modeling pipelines using Dockerized environments for reproducible experiments.
- Designing evaluation metrics and running large-scale experiments to assess explanation quality, interpretability, and robustness.

**Software Engineering Intern - Olfera**
**May 2025 to August 2025**

- Built and deployed backend infrastructure using AWS Amplify, integrating authentication and CI/CD pipelines across development, staging, and production environments to ensure reliable deployments.
- Designed 7+ normalized MySQL schemas for users, projects, experiments, and SOPs, reducing redundant queries by ~40% and improving data integrity.
- Developed secure GraphQL/AppSync API endpoints with role-based access control to protect sensitive research data and support scalable access patterns.

**Software Engineering Intern - Runopt**
**January 2025 to May 2025**

- Developed a 3D visualization framework in PyVista for civil site design, enabling engineers to simulate and analyze stormwater detention layouts in under 10 seconds per run.
- Implemented a knapsack-based optimization algorithm to select from 9+ UGDC models, improving storage-to-cost ratio while keeping solutions within 5% of budget constraints.
- Integrated OpenMDAO into the optimization pipeline, reducing cross-module runtime by ~30% and improving scalability.

## Projects & Activities

**UC Berkeley IEEE Micromouse - Autonomous Robotics (PCB Engineering + Embedded Systems)**

- Engineered and soldered an autonomous maze-solving robot using encoder odometry, IMU data, and I2C communication.
- Implemented a real-time A* pathfinding algorithm for fast maze exploration and optimized turn decisions.

**Falling Walls Lab Finalist - German Center for Research and Innovation**

- Presented a data-analytics-driven platform concept supporting Bay Area small businesses, focusing on customer engagement and local discovery.
- Selected as a finalist among candidates from 60+ countries, representing innovation in community-focused tech.